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A miniaturized electronic nose with artificial neural network for anti-interference detection of mixed indoor hazardous gases

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A R T I C L E I N F O

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A B S T R A C T

Indoor air quality attracted great attention for its significant threats to human health and safety, especially the potential hazardous gases in kitchens. To meet the requirements of the anti-interference detection of multiple combustible gases, in this paper, a miniaturized electronic nose was developed using MOS sensor array for semi-quantitative and anti-interference detection of carbon monoxide and methane with the interference of hydrogen and formaldehyde. The sensor array was constructed using 6 MOS sensors and cross-reaction to target and interference gases. To implement the anti-interference capability, different models were utilized and evaluated including PCA, LDA and BP-ANN. The 10-fold cross validation results indicate that BP-ANN models have the best performance than other models with the accuracy of 93.35 % for CO and 93.22 % for CH_4 without interference. With the interference of H_2 and CH_2O, the BP-ANN model shows the accuracies of 78.92 % for CO and 89.75 % for CH_4. Adding interfering samples of H_2 has a more significant impact on BP-ANN models than adding that of CH_2O. The results demonstrate that the proposed e-nose with the BP-ANN model can realize semi-quantitative, simultaneous and anti-interference detection of CO and CH_4 in the interference environment, which provides a promising platform for gas sensing with multiple interference.

1. Introduction

Air quality is an important indicator to evaluate whether the environment is livable or not. Indoor combustible gases damage human health more directly and severely which mainly include carbon monoxide (CO) and methane (CH_4). They can cause extremely adverse effects on human beings such as physical discomfort, asphyxia, even serious fires and explosions. At least 250 million people use combustible gases in kitchens, more than 70,000 among them are seriously affected by asphyxia, poisoning, fires and explosions every year. In addition to the alarm functions that are currently achievable [1], the detection of mixed gases in a complex environment remains a huge challenge, especially for cross-reactive sensors. Therefore, it is of great necessity to research on a semi-quantitative and anti-interference detection method for mixed harmful gases.

In recent decades, there has been a rapid growth in the field of gas sensors and a variety of sensors have emerged such as semiconductor sensors [2], combustion-type sensors [3], solid-electrolyte type sensors [4], electrochemical sensor [5], optical sensors [6,7], acoustic sensors [8,9] and olfactory-based biosensors [10]. Among them, metal oxide semiconductor (MOS) sensors have been widely used in the field of gas detection due to their low cost, small size, easy fabrication, high sensitivity and long service life [11]. Although extensive efforts [12–15] have been made to develop MOS sensors with excellent performance, it remains a huge challenge to identify several gases based on a single sensor without other sensing technologies [16]. In general, the multisensory array was much more effective to construct an e-nose for the discrimination of gases. Compared with the most common detection method gas

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chromatography-mass spectrometry (GC–MS) [17–19], the multisensory array is more inaccurate as well as more susceptible to interfering gases. However, such methodology is ideal for rapid detection in practical applications as the results can be obtained in minutes. With the help of pattern recognition, recent studies have developed a great number of specific and intelligent e-nose systems for applications including toxic gases detection [20], air quality management [21], explosive sensing [22], food freshness and quality prediction [23,24], fire detection [25], agricultural monitoring [26–28], disease diagnosing [29,30], pipeline leak detection [31] and others [32].

A few years ago, a metal oxide semiconductor (MOS) sensor selection method using a pattern recognition technique [33] was proposed for the detection of multiple kinds of indoor air contaminants. The results showed that kernel-based principal components analysis (PCA) was capable to classify several indoor contaminants with low concentrations. Lin [34] developed a semiconductor gas sensor array for volatile organic compound gas mixture classification based on ELM, BP and SVM. The results showed that, compared to BP neural networks and SVM, ELM networks can achieve faster training speed and satisfactory classification accuracy. In 2017, a gas concentration estimation method based on a multivariate relevance vector machine (MVRVM) is proposed for mixed gas detection by using MOS gas sensor array [35]. The experimental results of the mixture showed that the average relative errors of CO and CH_4 are 5.58 % and 5.38 % both between the concentration range of 200−1200 ppm. Another similar study [36] developed a SnO_2 sensor array to detect pure target gases CO and CH_4. Finally, the LSR estimator achieved 95.5 % and 94.4 % minimum accuracies for 50−1000 ppm CH_4 and 10−200 ppm CO, respectively. Lately, five different MOS sensors were fabricated and formed a multiarray e-nose which could distinguish seven hazardous analytes based on PCA pattern analysis [37].

There are also some studies of anti-interference gas detection. In 2019, a special method [16] combined with a single MOS sensor with ultrasound was proposed for gas identification. The identification accuracies of 8 gases could be all 100 % without complex feature extraction. It demonstrated that increasing the ultrasonic strength could significantly improve the anti-interference capability. Moreover, an e-nose [38] with 14 MOS sensors was developed to distinguish 5 flammable liquids with the interference of mosquito repellent, perfume and hair jelly. This study collected 25 samples for each kind of flammable liquid and 15 interfering test samples. The results showed that BP-ANN could perfectly distinguish different flammable gases qualitatively with the average quantitative error of about 12 %.

However, few studies have realized the anti-interference identification of mixed multiple gases based on cross-sensitive sensors. Besides, commercial instruments in the market are not suitable for long-term and convenient air quality testing in daily life due to their few parameters, large size, high price or inconvenient operation.

In this paper, a semi-quantitative, simultaneous and anti-interference e-nose system based on MOS sensor array is proposed to detect 10−1000 ppm CO and 500−10,000 ppm CH_4 in the mixture (Fig.1). The system consists of three parts: gas blender module, detection module and concentration level classification module. The gas blender mixes different gases and provides required mixture for detection. All mixed gas samples are diluted by the standard air for the MOS sensor to work properly. When it is cleaned after detection, the standard air is directly discharged into the gas chamber. After that, the detection module collect the response of the sensor array to the mixed gases under the control of a microcontroller unit (MCU). CO and CH_4 are selected as target gases due to their flammability and risk. In addition, H_2 and CH_2O are used as interference gases to simulate a more complex environment. Finally, linear discriminant analysis (LDA), PCA and BP-ANN models are trained for pattern recognition based on responses of the sensor array.

2. Materials and methods

2.1. Materials

The gas blender was the GAS BLENDER 6000 produced by MCQ Instruments (Italy). The standard air (79 % nitrogen and 21 % oxygen), CO (3010 ppm, standard air diluted), CH_4 (30,100 ppm, standard air diluted), H_2 (1000 ppm, standard air diluted) were bought from Jingong Materials Co., Ltd (Hangzhou, Zhejiang). The CH_2O (150 ppm, standard air diluted) was bought from Best Gas Co., Ltd (Hangzhou, Zhejiang). The SHT20 sensor was bought from Sensirion Company (Switzerland).

2.2. Gas sensors

Since a single MOS sensor is cross-sensitive and poorly selective, making it difficult to distinguish between different gases. Thus it is necessary to obtain multiple responses to mixed gases employing a MOS sensor array. Here, six commercial MOS sensors are chosen to set up a sensor array considering the stability, low cost and easy accessibility. Other sensors such as electrochemical and optical sensors have better specific performance for gas detection, but they are usually expensive

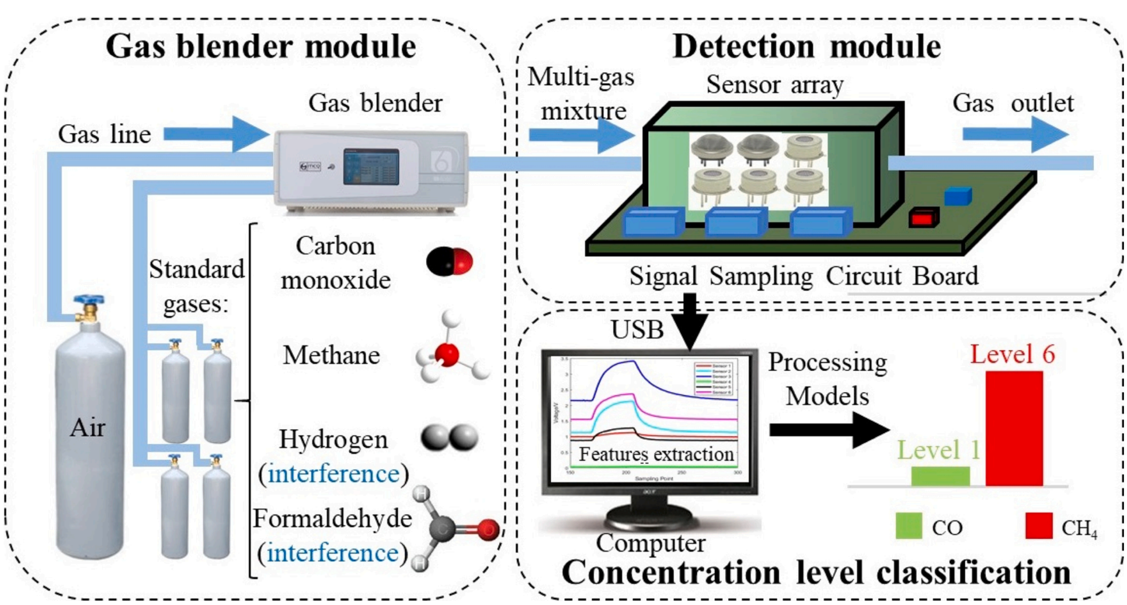

Fig. 1. The overall diagram of the electronic nose system.

and take up more space when forming sensor array. Therefore, they are not suitable for harmful gas monitoring in the domestic environment.

The MP-9, MP-4 and MP503 sensors used in this study were purchased from Winsen Electronics Technology Co., Ltd (Zhengzhou, Henan). And TGS821, TGS816 and TGS2602 were purchased from Figaro Engineering Inc (Japan). Table S1 in the supplementary material introduces several target gases of six sensors and the detecting range of some relevant gases in our study. All sensors are cross-sensitive to both target gases and interfering gases. So we will train data processing models to recognize and classify different target gases.

2.3. Fabrication of the electronic nose detection system

As mentioned above, the electronic nose system is made up of three parts: a gas blender module, a detection module and a concentration level classification module. It is developed to evaluate the concentrations of the mixed harmful gases using six MOS gas sensors and SHT20 for humidity and temperature monitoring. The Photograph of the electronic nose system is shown in Fig.S1 in the supplementary material. The whole system is placed in a fume hood except for gas cylinders. Here, we adopt a differential amplification and low-pass filter circuit to preprocess the initial sensor voltage signal and ensure that the voltage doesn't saturate during the experiment. Simultaneously, the signals are sent to the software on the computer for further processing and model training.

2.4. The classification of hazardous levels

According to the national indoor air quality standards and combustible gas detection standards of China, 10−1000 ppm and 500−10000 ppm are reasonable detection ranges of CO and CH_4, respectively. In order to reflect the classification result of the gas concentration more intuitively, we divide the target gas concentration into six levels in Table S2 in the supplementary material. H_2 and CH_2O are not detected as target gases, however, they are common and capable to interfere with sensor response. For H_2, concentrations are ranging from 5 to 100 ppm. For CH_2O, concentrations are ranging from 1 to 40 ppm. Limited to channels of the gas blender, we can only prepare a mixture of not more than three gases.

Table S3 in the supplementary material shows the experimental gas samples including some interfering samples. Samples of interfering gases don't cover all combinations of levels, but they evenly cover most of them with not more than one level interval. Finally, we collected a total of 411 gas samples of which 183 samples exclude interfering gases. Without doubt, we expect that the interfering gaseous samples have as little impact on the classification results of target gases as possible.

2.5. Experiment process

In this study, the actual concentrations of different target gases in the mixture were automatically calculated and provided by the gas blender. Considering that a low flow rate will cause the response time to be too long, while a high flow rate will result in a large amount of gas consumption. Therefore, we decided to test at a flow rate of 1000 mL/min. In order to eliminate the effects of temperature and humidity, we considered them in the algorithm model. The experiment process consists of several parts. Firstly, the sensor array was preheated for more than 30 min in advance and cleaned in standard air with the flow rate of 1000 mL/min for nearly 30 min until the voltage of each sensor was below the corresponding preset threshold. Next, the gas blender injected the mixed gas sample into the gas chamber at a constant flow rate for 5 min. In the meantime, features were extracted from the response of sensor array and calculated by the MCU in real-time. Finally, the sensor array was cleaned for 5 min. So it took just 10 min totally to detect one sample as well as cleaning. In addition, the detecting and cleaning process can be continuously cycled to detect more samples.

2.6. Features extraction and modelling methods

The original data of sensors were collected with a sampling frequency of about 0.3 Hz. Before establishing the concentration level discriminant model, we need to extract several features from the original data. First, we extracted the voltage baseline after cleaning. When the mixed sample entered the gas chamber, the sensor voltage rose and the maximum slope of increase (MSI) was extracted. The sensor's response reached its maximum response before cleaning the gas chamber and the maximum was used as a feature. Here we preprocessed the data so that these two feature values could be merged into resistance ratio R_i:

$$R_i = \frac{R_s}{R_o} = \frac{(V_{in} - P_i)/P_i}{(V_{in} - B_i)/B_i}$$

Where i stands for the ID of gas sensor ranging from 1 to 6, R_o is the resistance of the sensor S_i in dynamic standard air, R_s is the resistance of S_i when it finishes detecting, $V_{in} = 5\ V$ is the constant input voltage of the gas sensor, P_i is the maximum response of S_i during detection, B_i is the baseline of S_i before detection. We also calculated the peak area (PA) at the same time, which is the sum of the voltage during the detection minus the area of the baseline. So each sensor provides three features: MSI, R and PA. Although the temperature and humidity did not fluctuate too much, we still take them into consideration and regard them as two independent features. Finally, a total of 20 features of the sensor array are used for the training of all models. It is known to all that PCA [27,37,39], LDA [27,40] and ANN [41,42] are methods used commonly in the field of pattern recognition. PCA can be described as an 'unsupervised' algorithm but LDA is 'supervised'. However, their performance in this study is unknown. In some cases, the performance of them can even be better than ANN. Therefore, it is necessary to study the feasibility of these three widely used methods in this application. In this study, we used the classic calculation procedures of PCA and LDA to train models of CO and CH_4. Besides, 3-layer BP-ANN models were built for CO and CH_4 respectively. The models employed tan-sigmoid activation function and the scaled conjugate gradient (SCG) algorithm. The input layer contains 20 points representing 20 features, the middle layer contains 11 points and the output layer contains 6 points presenting 6 levels. If the loop reaches the maximum number of 1000 or the gradient falls below the minimum gradient of 10^{-6}, the training is stopped.

Data acquisition was conducted on LabVIEW software (professional edition, 2015). Feature extraction and model construction were implemented on MATLAB R2017b. The gas chamber simulation was performed using FloXpress in SOLIDWORKS (2016). Visualization of the results was achieved using GraphPad Prism 7.

3. Results and discussion

3.1. Fluid dynamics simulation of the gas chamber

To verify the rationality of the gas chamber structure, airflow simulation were implemented in SOLIDWORKS. The results of the airflow simulation are shown in Fig.S2 in the supplementary material. Fig.S2a is the 3D structure diagram of gas chamber with six grooves for the assembly of the sensor array. For convenience, we removed the top surface to observe the inside and added some red arrows indicating the airflow direction. The gas flows through all sensors' surfaces quickly indicating that it can fully contact each sensor whether during detection or cleaning (Fig.S2b). Then it enters the bottom through gaps and exits from the left outlet (Fig.S2c). Therefore, we proves that this gas chamber is reasonable, which allows the gas to contact sensor array effectively.

3.2. Stability of the sensor array

As is known to all, the sensor array heats up during operation, which will cause the temperature inside the gas chamber to be significantly higher than the outside. Besides, if the chamber is adhered and sealed on PCB, temperature and humidity will be relatively stable without severe fluctuations during experiments. If not, the external air comes into it and the humidity becomes abnormal. Thus, we recorded these two parameters of each sample inside the chamber.

The results in Fig. 2a proved that the temperature could remain stable for at least six months, ranging from 33 to 40 °C. Moreover, the relative humidity was also always below 4.2 % (Fig. 2b), which was equivalent to an extremely dry gas environment. The small fluctuations in both of them may be due to the influence of air conditioner and fume cupboard. The baseline stability of the sensor will affect the quality of data as well. Therefore, it is necessary to investigate the baseline stability of the sensor array. After being preheated, the sensor array was cleaned with standard air for 30 min at a flow rate of 1000 mL/min. We observed that after sending the air into gas chamber, each sensor's response dropped significantly and gradually approached a nearly steady state. For more than half a year, we measured the stable baseline response of each sensor for more than three times at different periods (Fig. 2c). Though baselines fluctuated, it is worth noting that most of them tended to be more stable. This phenomenon may be due to insufficient warm-up time for newly purchased sensors, which caused the previous baselines to be unstable. However, in most cases, fluctuation of each sensor was much smaller than its response to target gases and would not severely interfere with the classification of gases.

3.3. Gas sensor array calibration

Although the sensor's specifications were described in the sensor data sheet, the experimental condition in our study was changed observably and became more complex. Therefore, we must validate the performance of all gas sensors to ensure that their sensing functions are still reliable.

Firstly, we shown typical voltage-time responses of sensors when detecting different gases as well as cleaning gas chamber (Fig. 3). Fig. 3 shows typical voltage-time responses of sensors when detecting different gases as well as cleaning gas chamber. The high-purity air was utilized to dilute standard CH_4 gas into concentrations of 1000, 3000, 5000, 7000, 9000 ppm for calibration (Fig. 3a). Standard CO was diluted into 30, 100, 300, 600 and 900 ppm (Fig. 3b). H_2 was diluted into 5, 10, 20, 50 and 100 ppm (Fig. 3c). CH_2O was diluted into 7.5, 12, 15, 21 and 25 ppm (Fig. 3d). The sensor array continuously detected different concentrations of the same gas. It can be seen that the influence of the interfering gases on sensors is similar to that of target gases, which inevitably increases the difficulty of classification. The first batch of responses in Fig. 3c is abnormal compared to others because the gas blender tends to be unstable when it is preparing the lowest concentration of H_2 for the flow of H_2 is too small and difficult to be controlled accurately. However, this phenomenon was rare and H_2 was one of interfering gases, it had almost no effect on our results. On the whole, the baselines of sensors are stable and responses show a positive correlation with concentrations. These results demonstrate that sensors' performances are normal, and the sampling process is feasible.

Secondly, we evaluated linearities of the sensor array to CO and CH_4. The high-purity air was utilized to dilute standard CH_4 gas into concentrations of 1000, 3000, 5000, 7000, 9000 ppm for calibration. CO was diluted into 100, 300, 600 and 900 ppm. Several repetitive tests were implemented at each concentration to study the sensitivity and repeatability of sensors. To eliminate the effects of baseline drift, we subtracted baseline from the maximum sensor response in calibration curves. When exposed to CH_4, the response of S1 increases about 1.47 V from 1000 ppm to 9000 ppm with sensitivity of 1.84×10^{-4} V/ppm (Fig. 4a). Under the same conditions, responses of S2 to S6 raise 0.765, 0.742, 0.262, 1.45 and 0.06 V (Fig. 4b-f) with sensitivities of 0.95625, 0.9275, 0.3275, 1.8125, 0.075×10^{-4} V/ppm, respectively. All gas sensors have good linearities ($R^2 > 0.95$) to CH_4. Moreover, we can preliminarily conclude that S1 and S5 are more sensitive to CH_4. When exposed to CO, the response of S1 increases about 1.208 V in the range of 100 ppm–900 ppm with sensitivity of 1.51×10^{-3} V/ppm (Fig. 4g). Besides, responses of S2 to S6 raise 1.121, 1.310, 0.149, 0.182 and 0.182 V (Fig. 4h-l) with sensitivities of 1.40125, 1.6375, 0.18625, 0.2275, 0.2275×10^{-3} V/ppm, respectively. Although linearities of CO is a bit lower than that of CH_4, they are still acceptable ($R^2 > 0.88$). Poor linearity of S2 may be due to the narrow linear detection range for CO. It is obvious that S2 and S3 are more sensitive and repeatable to CO. The input voltage of S1 is 5 V. Thus S1 does not work in low temperature mode to detect CO, which may account for the poor repeatability of S1. Besides, S4 and S5 are not sensitive to CO for their response are quite small, leading to unstable response. The above discussion proves that these gas sensors have good detection performance for both CH_4 and CO. Different sensitivities are helpful for later identification of different components in mixed gas.

3.4. Recognition and classification of mixed gases

In order to implement and test the classification ability of this system, we trained different models utilizing three methods: LDA, PCA and BP-ANN. Then, we performed concentration level classification for different sample sets. We randomly divided each sample set into two parts, training sets containing 90 % data and testing sets containing the rest. Then we use PCA and LDA methods to obtain classification results of different training sets

In Fig. 5, 183 samples with only target gases and their mixture were used as the sample set. As is shown in Fig. 5a, LDA models can effectively distinguish different levels of two target gases without interfering gases. However, the sum of the two principal components can only contain nearly 75 % information of the original data, so the classification result of PCA is unacceptable. It is noticed that PCA models show a bit more overlapped in Fig.5b for low levels of CO and high levels of CH_4 leading to a decrease in overall accuracy. This may be due to the actual detection concentration is close to the upper or lower detection limits of the sensor, causing the sensor to be insensitive at lower concentration levels and too saturated at higher concentration levels. There is no doubt that these two extreme parts will increase the difficulty of classification.

In Fig. 6, all 411 samples were used as the sample set. After adding

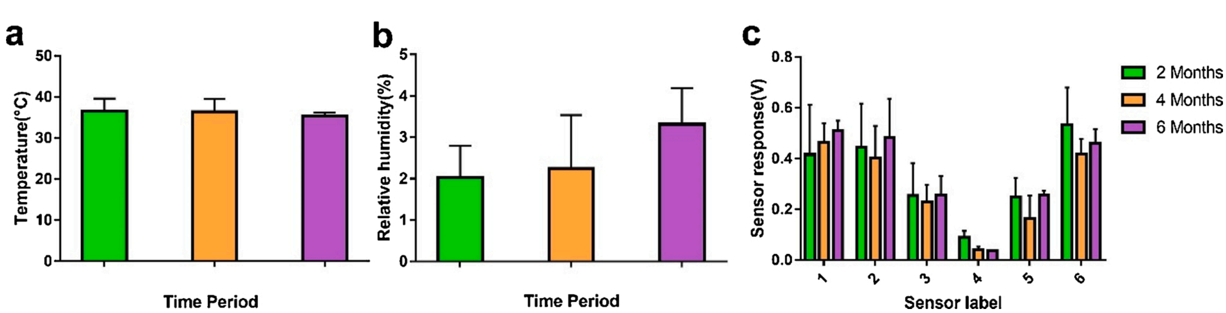

Fig. 2. Evaluation of sensor array long-time stability every two months: (a) Temperature; (b) Relative humidity; (c) Baselines of gas sensors.

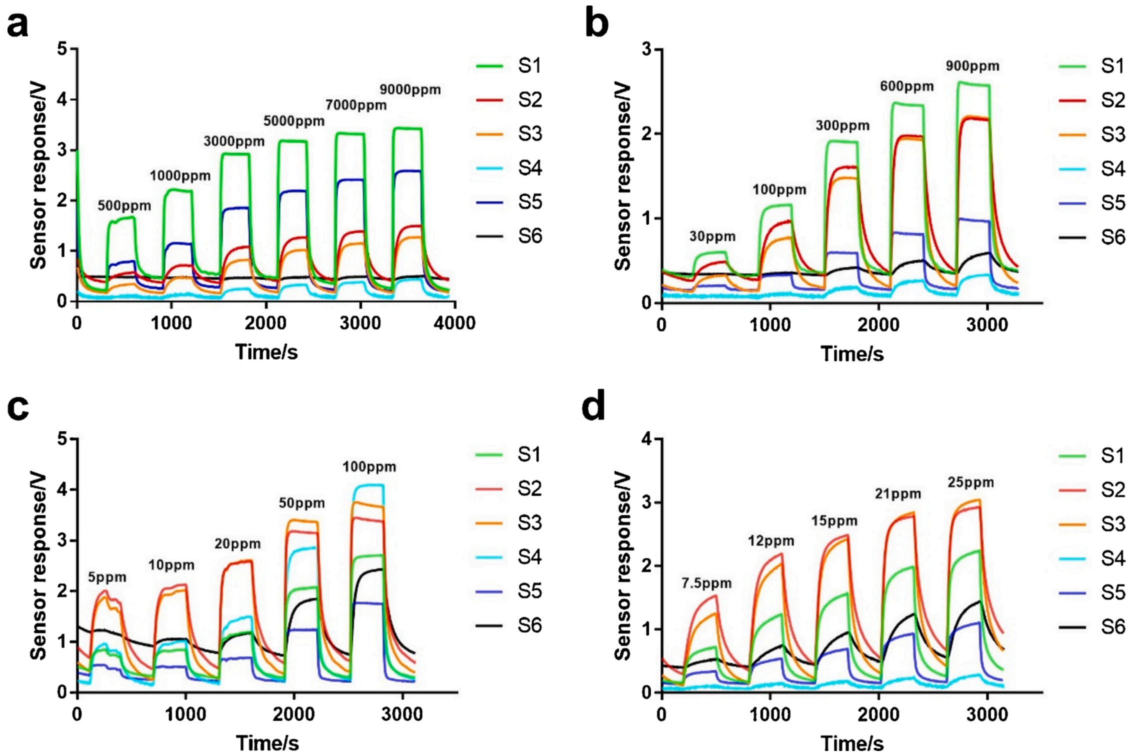

Fig. 3. The voltage-time response of the gas sensor array: (a) CH_4; (b) CO; (c) H_2; (d) CH_2O.

interfering samples, both of them are greatly influenced. Although a few points are crowded and difficult to be distinguished in high levels, LDA models can still tell the distance between most levels (Fig. 6a). Not surprisingly, the results based on the PCA method are ineffective and severely overlapped, especially for CO (Fig. 6b). This means that the model cannot accurately distinguish the level of CO, which causes a high probability of misjudgment. But it is still effective when distinguishing low levels of CH_4.

In order to verify the average classification accuracy of the detection system, we performed 10-fold cross validation on three kinds of models for two cases: no interfering samples and adding all interfering samples. Table S4 in the supplementary material shows the distribution of target and interfering gaseous samples for each case in detail. The proportion of interfering gaseous samples in each case is large enough and samples are nearly evenly distributed. The concentration level classification results are shown in Table 1. It can be found that the BP-ANN shows the best accuracy of up to 93.35 % for CO and 93.22 % for CH_4 in Case 1. Under the same condition, the performance of the LDA method is quite close to the BP-ANN. However, the performance of the PCA method is the worst with the accuracy of only 40.44 % for CO and 35.52 % for CH_4. In Case 2, the BP-ANN model still has complete advantages compared to PCA and LDA with the accuracies of 78.92 % for CO and 89.75 % for CH_4. The results of the PCA method are still unsatisfactory. Besides, the LDA model doesn't show outstanding accuracy as well and there is a significant difference from the accuracy of BP-ANN. Since each level is represented by several specific concentration values perhaps sampled only once or twice, which puts a high demand on the model's self-learning ability. Fortunately, the ANN method with strong self-learning ability enables the model to classify the concentration level of untrained samples. It is beneficial to enhance the anti-interference ability of the model.

Moreover, we conducted two more cases (Case 3 and Case 4) using BP-ANN so as to further figure out which kind of interfering samples significantly reduces the accuracy. In Case 3, we added samples containing CH_2O but no H_2 to the sample set. Similarly, samples containing H_2 but no CH_2O are added to Case 4. It should be noted that the amount of interfering samples were only 46 in Case 4 which was obviously insufficient compared to target gas samples. Thus we divided target gas samples and interfering samples into ten equal parts respectively and ensured that the ratio of H_2 samples in the training set was invariant. Whether interfering samples were in training set or testing set, they were duplicated 3 times to 184 totally which is very close to the amount of target gas samples. This method effectively increases the number of interfering samples. As mentioned above, the average accuracy is also calculated by 10-fold cross-validation. Interestingly, the addition of 133 CH_2O interfering samples has almost no effect on the accuracy of the BP-ANN model in Case 3. The classification accuracy rates of CO and CH_4 are 88.52 % and 93.85 % respectively. The effect of adding H_2 samples is clearly higher than that of adding CH_2O samples with the accuracy of 69.65 % for CO and 83.24 % for CH_4. It is apparent that the classification accuracy of CO is worse than that of CH_4 in most cases. This may be related to the smaller detection range of CO, which makes it more difficult to accurately recognize levels especially in an interference environment.

4. Conclusions

There is a great demand for the detection of harmful gases indoors, especially CO and CH_4. But in complex environments with multiple targets and multiple interferences, non-specific sensing technologies often encounter great difficulties. This study presented an e-nose detection system for semi-quantitative, simultaneous and anti-interference detection of mixed harmful gases. We tried to detect at least one target gas in the presence of up to two interfering gases H_2 and CH_2O. Depending on the MOS sensor array and data processing models, the proposed system can measure two target gases including CO and CH_4 which directly relate to human health and safety. Detection ranges of target gases were determined according to industry regulations. Next, each concentration range was divided into six levels. In order to verify the classification accuracy of the detection system, we adopted LDA, PCA and BP-ANN methods to establish models and evaluate them using 10-fold cross-validation.

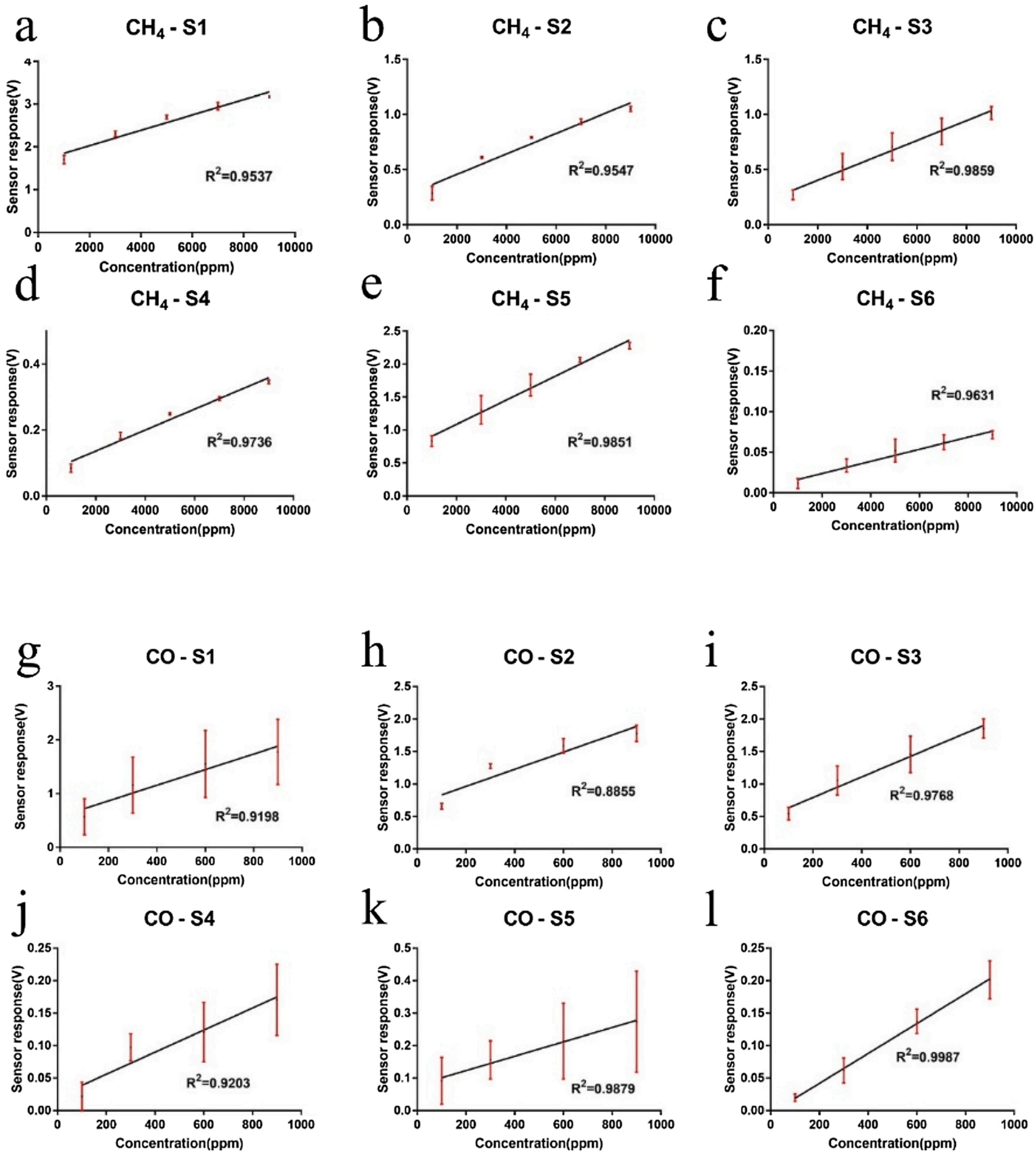

Fig. 4. Calibration curves of six gas sensors with CH_4: (a) S1; (b) S2; (c) S3; (d) S4; (e) S5; (f) S6; with CO: (g) S1; (h) S2; (i) S3; (j) S4; (k) S5; (l) S6.

The results prove that the model trained by BP-ANN has the best performance for both target gases whether there are interfering samples or not. Besides, LDA is more suitable and anti-interference for classification and classification than PCA in this application. In the absence of interfering gas samples, the performance of LDA can even be close to that of BP-ANN. Moreover, the BP-ANN models present better anti-interference performance for CH_2O than H_2. In the next stage, we will study how to detect more gases simultaneously, shorten detection time and reduce the impact of multiple interferences without reducing accuracy. More self-learning models will be explored to optimize the e-nose system.

CRediT authorship contribution statement

Junyu Zhang: Conceptualization, Investigation, Data curation, Writing - original draft, Writing - review & editing, Visualization. **Yingying Xue:** Investigation, Methodology, Software, Data curation, Writing - review & editing. **Qiyong Sun:** Conceptualization, Resources, Supervision, Funding acquisition. **Tao Zhang:** Resources, Investigation, Writing - review & editing. **Yuantao Chen:** Resources, Validation, Writing - review & editing. **Weijie Yu:** Resources, Writing - review & editing, Funding acquisition. **Yizhou Xiong:** Resources, Writing - review & editing, Funding acquisition. **Xinwei Wei:** Supervision, Visualization, Writing - review & editing. **Guitao Yu:** Supervision, Writing - review & editing, Funding acquisition. **Hao Wan:** Conceptualization, Writing - original draft, Writing - review & editing, Supervision, Project

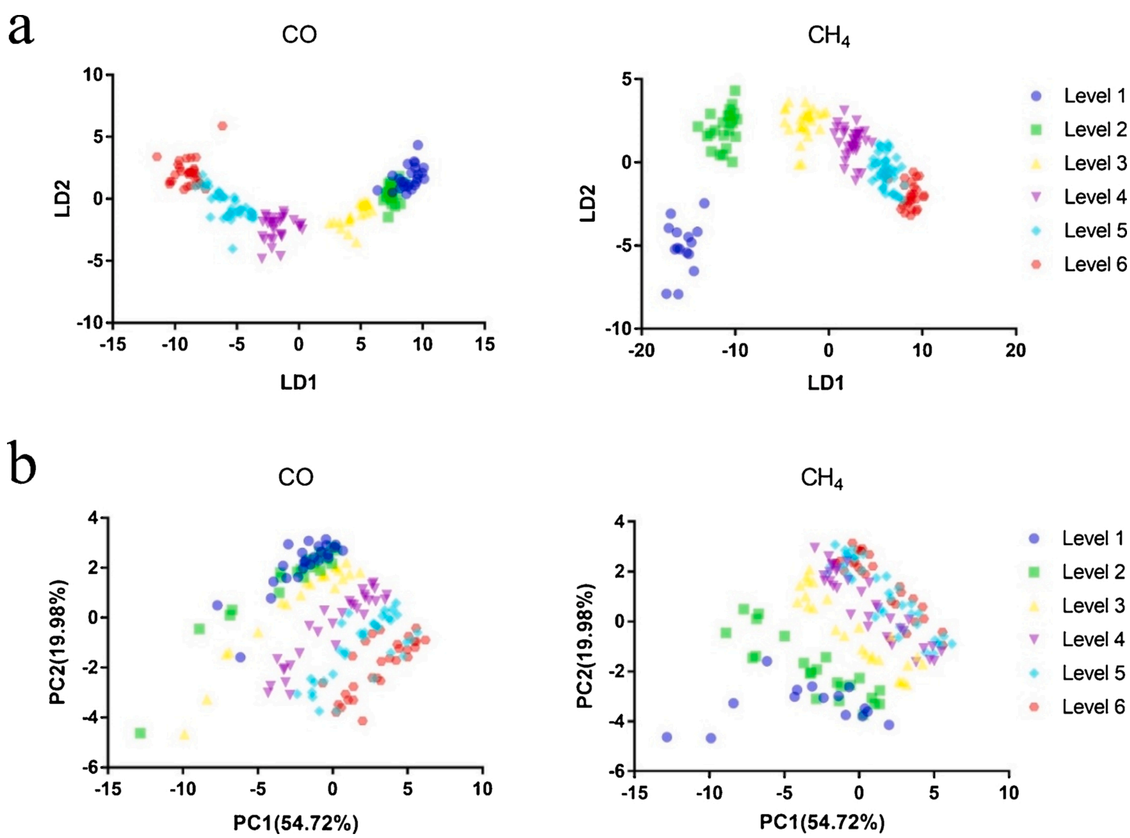

Fig. 5. Concentration level classification of CO and CH_4 without adding interfering samples: (a) LDA models; (b) PCA models.

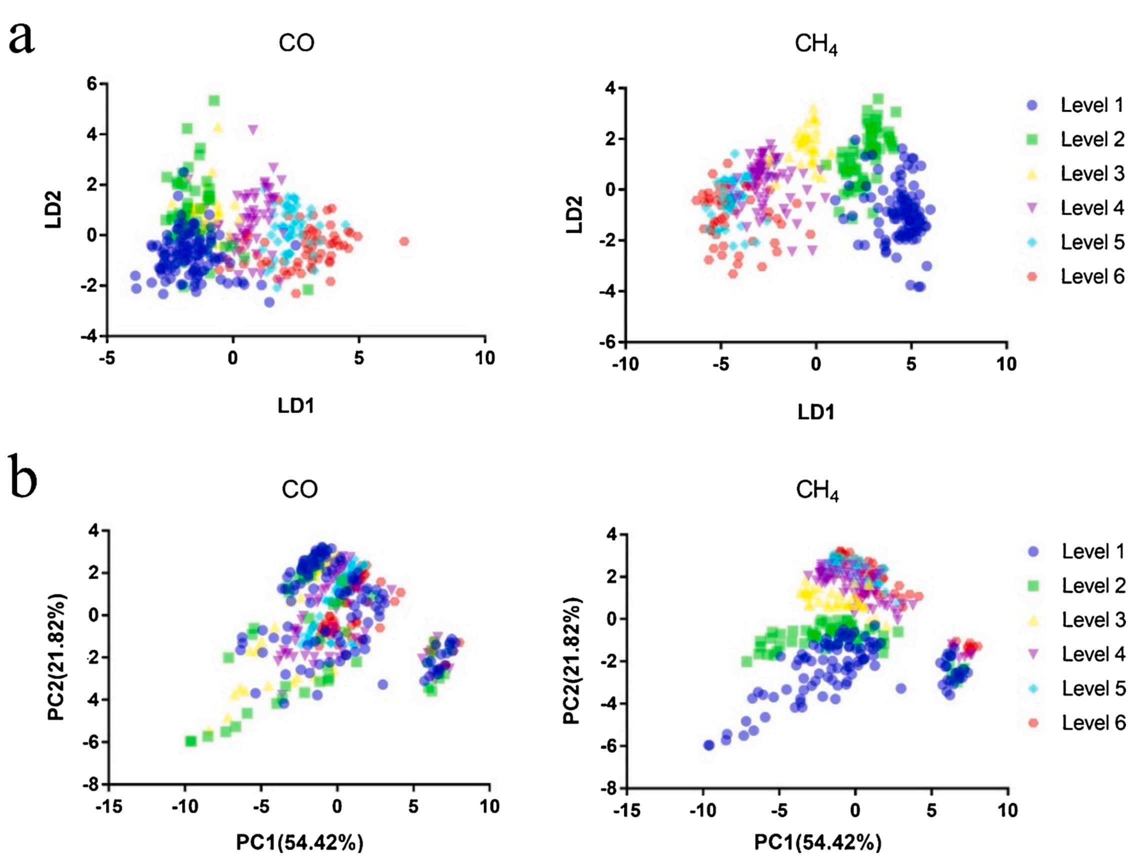

Fig. 6. Concentration level classification of CO and CH_4 after adding all interfering samples: (a) LDA models; (b) PCA models.

Table 1
Concentration level classification results of the detection system.

	Case 1			Case 2			Case 3	Case 4
	LDA	PCA	BP-ANN	LDA	PCA	BP-ANN	BP-ANN	BP-ANN
CO	89.07 %	40.44 %	93.35 %	51.22 %	20.92 %	78.92 %	88.52 %	69.65 %
CH_4	92.35%	35.52 %	93.22 %	73.17 %	47.69 %	89.75 %	93.85 %	83.24 %

Where Case 1: no interfering samples (183); Case 2: adding all interfering samples (183 + 228); Case 3: adding CH_2O interfering samples only (183 + 133); Case 4: adding H_2 interfering samples only (183 + 46*4).

administration, Funding acquisition. **Ping Wang:** Conceptualization, Supervision, Writing - review & editing, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.snb.2020.128822.

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